

Event Driven State Machine Lab

BME554L - Fall 2026 - Palmeri

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Tip

I would recommend doing this with pencil and paper first, not immediately jumping into a tool. Err on the side of more detail than not.

Digital Kitchen Timer

Generate a state diagram for a simple digital kitchen timer that has the following buttons:

- Start/Stop Button
- Reset Button
- Pause Button
- Up/Down/Left/Right Arrow Buttons (4)

The timer outputs time intervals in the format: HH:MM:SS

Functional States

- The kitchen timer has an “Idle” state when it isn’t actively counting down time. In this state:
 - The Up/Down buttons increment the time for the HH, MM or SS values.

- The **Left/Right** buttons let you move between **HH**, **MM** and **SS**.
- Time intervals will wrap around to 00 when a reasonable max value is exceeded.
- The **Reset** button sets all of the time values to 00.
- The **Pause** button is inactive.
- **Start/Stop** will start the timer countdown when in **Idle**.
- In the **Countdown** state:
 - The **Start/Stop** button will stop and reset the countdown interval.
 - The **Pause** button will pause, but not reset, the countdown time.
- When the countdown is done, the timer enters an **Alarm** state, where:
 - The **Up/Down/Right/Left** arrow buttons, **Reset** and **Pause** buttons are inactive.
 - **Start/Stop** will stop the alarm and return the timer to **Idle** with the previously set countdown time.

Tip

If any functional state expectation isn't explicitly described above, please implement something reasonable.

Make sure that your state diagram:

- Has initialization (i.e., batteries put in) and termination (i.e., completely powered off) dots.
- Includes all functional states, including (but not limited to):
 - **Idle**
 - **Countdown**
 - **Paused**
 - **Alarm**
- Consider if states should have entry and exit actions.
- Make sure all transitions between states are annotated with associated triggering events.

Generate the State Diagram

Formally render your state diagram using one of the tools outlined on the [State Diagram Tools](#) page.

Tip

For PlantUML-generated digrams, if the diagram's layout is difficult to follow using the automated layout, consider manually specifying relative positions and

length of arrows using the following annotation: <https://crashedmind.github.io/PlantUMLHitchhikersGuide/layout/layout.html#::~text=6.>Tip.

Gradescope

Upload your rendered state diagram to the associated Gradescope assignment.